Question1: MSQ

(15) Minutes [ ] / 10 Marks

1	2	3	4	5	6	7	8	9	10

1. **How many passes does an insertion sort algorithm consist of?**
   A. N
   B. N-1
   C. N+1
   D. N2

2. **Breadth First Search graph algorithm uses ——— data structure for its implementation:**
   A. Queue
   B. Stack
   C. Tree
   D. All the above answers are incorrect

3. **The following array is a max heap: [10, 3, 5, 1, 4, 2].**
   A. True
   B. False

4. **Which sorting algorithm will take least time when all elements of input array are identical?**
   A. Heap sort
   B. Merge sort
   C. Insertion sort
   D. All the above answers are correct

5. **If f(n) = O(g(n)) and g(n) = O(f(n)) then f(n) = g(n)**
   A. True
   B. False

6. **To apply dynamic programming on a computational problem, it must have two key properties: optimal substructure and:**
   A. It has overlapping subproblems.
   B. It has multiple subproblems.
   C. It contains no overlapping subproblems.
   D. It consists of a single problem.
   E. All answers are incorrect.

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7. **A problem that can be solved in $O(n^c)$ time, where c is a positive number, belongs to the set of:**
   A. NP-Complete problems.
   B. NP problems.
   C. P problems.
   D. Problems solvable in exponential time.
   E. All answers are incorrect.

8. **A computational problem in which the aim is to find a solution with the best possible score, according to some objective function, is:**
   A. A decision problem.
   B. A polynomial problem.
   C. An NP-Complete problem.
   D. An NP problem.
   E. All answers are incorrect.

9. **The objective of the knapsack problem is:**
   A. To get minimum total value in the knapsack.
   B. To get maximum weight in the knapsack.
   C. To get minimum weight in the knapsack.
   D. To get maximum total value in the knapsack

10. **If f(n) = 5nlogn and g(n) = n log 5n then f = Θ(g)**
   A. True
   B. False

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Question 2 **Fundamentals of Algorithms Analysis** **(10) Minutes [ ] / 6 Marks**

1. **For each pair of functions f(n) and g(n) in the table below, write O, Ω, or Θ in the appropriate space, depending on whether f(n) = O(g(n)), f(n) = Ω(g(n)), or f(n) = Θ(g(n)). If there is more than one relation between f(n) and g(n), write only the strongest one. The first line is a demo solution. We use lg to denote the base-2 logarithm. [3 marks]**

	n	$n \lg n$	n^2
$n \lg^2 n$	Ω	Ω	O
$2^{\lg^2 n}$			
$\lg(n!)$			
$n^{\lg 3}$			

2. **Find Θ , c_1, c_2, n_0 of $f(n)=(n^2+\sqrt{n})$ [3 marks]**

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Question 3 **Recurrence** **(15) Minutes [ ] / 7 Marks**

Use the iteration and Master methods to give the tight upper bound (big-O) to the following recurrence:

$4T(n/2) + n^2$

1. **Iteration method: [3 marks]**

2. **Master method [2 marks]**

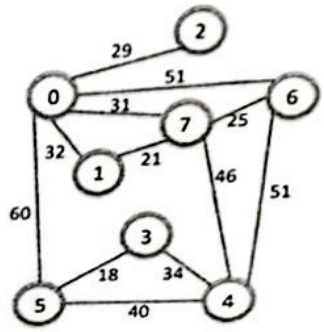
3. ***Determine the complexity* of the following code in Big-Oh notation (Hint: use *Recursion Tree method*). [2 marks]**

```
foo( n ){
 if (n == 0) return;
 foo( n – 1);
 foo( n – 1);
 print n;
}
```

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Question 4 **Graph** **(30) Minutes** **[ ] / 12 Marks**

1. **Use undirected-weighted graph below to find a minimum spanning tree (MST) by using *Kruskal algorithm*. (1) draw the resulting minimum spanning tree, (2) specify the value of MCST and (3) specify the order in which edges get added to the spanning tree?**



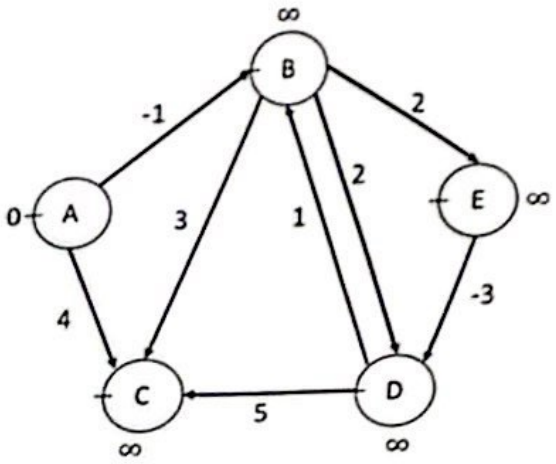
A. **The resulting minimum spanning tree:** [1 mark]

B. **The value of MCST:** [ ] [1 mark]

C. **The order in which edges get added to the spanning tree:** [2 marks]

D. **If Prim's algorithm is applied, what is the order in which vertices get added to the spanning tree (i.e., removed from the priority queue)? Start with the vertex 0.** [2 marks]

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A. Which algorithm you have to use to find the shortest path from vertex A to each other vertex in the above graph? [1 mark]

a) Dijkstra's Algorithm
b) Bellman-ford Algorithm

B. Why did you select this algorithm? [1mark]

C. Fill the table below with the shortest paths from vertex A to each other vertices in the graph? [3 mark]

From A to	Path	Cost
C		
D		
E		

D. In the above graph, write the in-degree of all vertices. [1 mark]

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Question 5 **Dynamic programing** **(20) Minutes** **[ ] / 5 Marks**

1. **Consider the following 4 items and their weights and values:**

Item	Liter	Lifetime
Item 1	10	8
Item 2	200	14
Item 3	50	2
Item 4	20	80

Assume you want to buy the items with the highest lifetime, such that you can take a whole item or part of it. Also, assume you have a limited capacity, i.e., you cannot exceed 100 Liters.

A. **What does this problem resemble (i.e.., what is the name of this problem)? [1 mark]**

B. **What type of an algorithm that is used to solve this problem? [1 mark]**

C. **Apply the algorithm you mentioned in (B) to solve the problem. Your solution must show which items you have selected, and how many liters for each item, and the maximum lifetime of the items you selected. [3 marks]**

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